PROJECT REPORT

Python Hackathon

Problem Formulation

The problem being solved:

Local and small scale farmers often waste a lot of crop yield due to inconsistency in being able to predict the future weather pattern, soil info, and lack of experience. The “what if that happened” leads to inefficiency in the productive output which hinders the sustainable growth especially at a crucial time like now.

That is why we will develop GreenTwin, a digital twin farm decision engine, which lets the user build their farm plot with their specific feature and question an AI about any simulatable event. Then the AI will process that at the backend and get back with a concise answer to what the output of the simulation was and any helpful feedback.

Agriculture need & Justification:

- Efficient Yield: to have a productive yield
- Better planning: to reduce inefficiency
- Climate resilience: to prevent unexpected failure

Target Farming Context:

Target Users: local, inexperienced, small scale farmers

Initial setting: A farm represented digitally

Input Data: Digital farm

Output Data: Digital farm view; simulation summary; helpful guidance

Operational Constraint:

- Limited access to live data
- Difference in crop, weather, and local environment

Project Constraint:

The initial MVP will have a web interface with a digital farm as context for the AI that will be able to call different simulation scenario at the backend and give a summary of that

Literature Survey

Agriculture and green energy are increasingly being integrated with AI technology because of the advanced chain of logic that it will provide. Prior research suggests that a digital twin can extend conventional monitoring and decision making with a virtual farm and environment where one can foresee conditions before actually {} at the farmland. However the extent of the effectiveness of the digital twin engine will depend on the type and quality of data.

Research/Journal Papers that address the problem:

- 1) Digital Twins in Agriculture: A State-of-the-art Review by Purcell and Neubauer
- 2) Digital Twins in Agriculture: A review of recent progress and open issues by Wang
- 3) Digital Twins in Agriculture: Orchestration and Applications by Escribà-Gelonch et al. (2024)
- 4) Current applications and potential future directions of reinforcement learning-based Digital Twins in agriculture
- 5) Food and Agriculture Organization of the United Nations (FAO)

Research Gaps, Objectives & SDG Alignment:

Research Gap at existing work:

- Data Integration: all the data might exist in different systems
- Accessibility: Advanced digital twins may need advanced sensors and infrastructure
- Actional Recommendation: Simulating a scenario does not give a decision
- Sustainability Evaluation: Looking at productive efficiency may overlook sustainability

Objective:

- 1) Digital farm representation: Build a digital view of one sample farm, including its layout, selected crop and relevant soil or environmental attributes.

- 2) Data-driven recommendations: Implement at least three explainable decision rules related to irrigation or resource management, using clearly defined input thresholds and crop-specific assumptions.
- 3) What-if analysis: Enable users to modify at least two input conditions and compare the resulting recommendations against the original scenario.
- 4) Resource-awareness indicators: Display at least two indicators, such as estimated irrigation requirement and a water-use category, with assumptions clearly documented.
- 5) Functional validation: Test at least five predefined scenarios, including normal, dry and wetter conditions, and verify that the system produces the expected output for each test case.

8. Sustainable Development Goal (SDG) Alignment

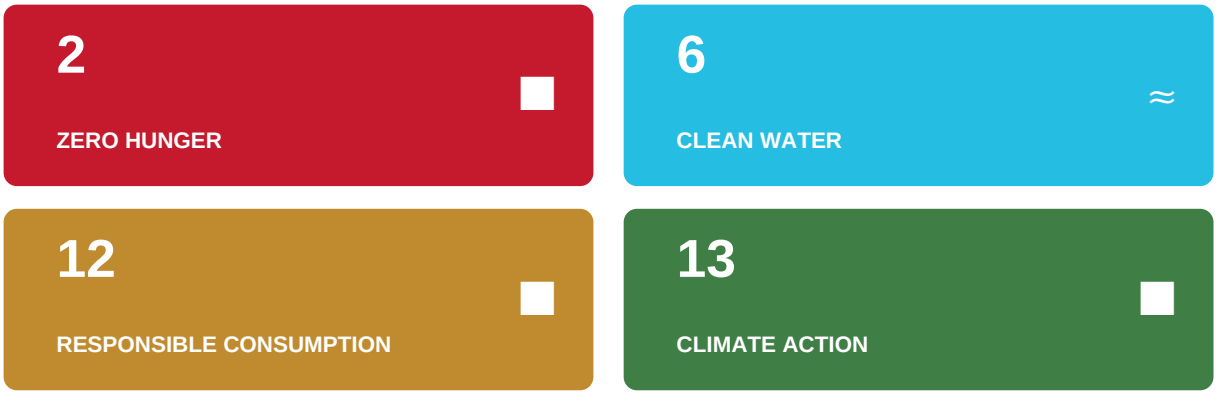

Figure 2. The project's main SDG connections. The coloured panels are illustrative and are not official UN SDG icon artwork.

8.1 SDG 2: Zero Hunger

The project supports sustainable agriculture by helping farmers assess crop-related scenarios, improve farm planning, and potentially reduce avoidable crop losses. It aligns most closely with Target 2.4, which focuses on sustainable food production systems and resilient agricultural practices.

8.2 SDG 6: Clean Water and Sanitation

By enabling users to explore irrigation and water-availability scenarios, the system can support more efficient water-management decisions. It aligns with Target 6.4 on water-use efficiency and sustainable freshwater withdrawals.

8.3 SDG 12: Responsible Consumption and Production

Scenario comparisons can encourage more responsible use of water and other agricultural inputs. This supports Target 12.2, which calls for sustainable management and efficient use of natural resources.

8.4 SDG 13: Climate Action

The system can help users explore the potential effects of climate-related conditions, such as reduced rainfall or higher temperatures, and consider adaptive actions. This supports Target 13.1 on strengthening resilience and adaptive capacity to climate-related hazards.